MATH 152 **QUIZ 4** Name Solution

Mrs. Bonny Tighe 7.2, 7.3

25 points TA_____Wed 9/25/2019

1. Evaluate the integrals.

3pts a) $\int t\,(\sin^2 t)\,dt$

$u = t \quad dv = \sin^2 t = \frac{1}{2} - \frac{1}{2}\cos 2t\,dt$

$du = dt \quad v = \frac{1}{2}t - \frac{1}{4}\sin 2t$

$t\left(\frac{1}{2}t - \frac{1}{4}\sin t\right) - \int\left(\frac{1}{2}t - \frac{1}{4}\sin 2t\right)dt$

$t\left(\frac{1}{2}t - \frac{1}{4}\sin t\right) - \frac{1}{4}t^2 - \frac{1}{8}\cos 2t + C$

4pts b) $\int \sec^3(3x)\tan^3(3x)\,dx = \int \sec^2 3x \tan^2 3x \underbrace{\sec 3x \tan 3x\,dx}_{du}$

$\int \sec^2 3x\,(\sec^2 3x - 1)\sec 3x \tan 3x\,dx$

$u = \sec 3x$

$du = 3\sec 3x \tan 3x\,dx$

$\frac{du}{3\sec 3x \tan 3x} = dx$

$\frac{1}{3}\int u^2(u^2 - 1)\,du$

$\frac{1}{3}\int (u^4 - u^2)\,du$

$\frac{1}{3}\cdot\frac{1}{5}u^5 - \frac{1}{3}\cdot\frac{1}{3}u^3 + C$

$\frac{1}{15}\sec^5(3x) - \frac{1}{9}\sec^3(3x) + C$

6 pts c) $\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}\,dx$

$x = 2\sin\theta$ | $\sin\theta = x/2$

$dx = 2\cos\theta\,d\theta$ | $\sin\theta = \sqrt{3}/2$, $\theta = \pi/3$

$\sin\theta = 1/2$ $\theta = \pi/6$

$$\int_{\pi/6}^{\pi/3} \frac{\sqrt{4-4\sin^2\theta}}{4\sin^2\theta}\, 2\cos\theta\,d\theta = \int_{\pi/6}^{\pi/3} \frac{\sqrt{4(1-\sin^2\theta)}\; 2\cos\theta\,d\theta}{4\sin^2\theta}$$

$$\int_{\pi/6}^{\pi/3} \frac{2\cos\theta\, 2\cos\theta\,d\theta}{4\sin^2\theta} = \int_{\pi/6}^{\pi/3} \cot^2\theta\,d\theta$$

$$\int_{\pi/6}^{\pi/3} (\csc^2\theta - 1)\,d\theta = -\cot\theta - \theta \Big|_{\pi/6}^{\pi/3}$$

$$(-\cot\pi/3 - \pi/3) - (-\cot\pi/6 - \pi/6) = -\sqrt{3}/3 - \pi/3 + \sqrt{3} - \pi/6 \text{ or}$$

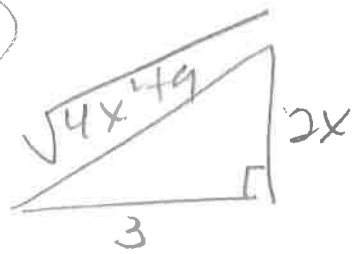
$$2\sqrt{3}/3 - \pi/6$$

5 pts d) $\int \frac{1}{\sqrt{4x^2+9}}\,dx$

$x = 3/2\tan\theta$

$2x = 3\tan\theta$

$dx = 3/2\sec^2\theta\,d\theta$



$$\int \frac{3/2\sec^2\theta\,d\theta}{\sqrt{9\tan^2\theta+9}} = \int \frac{3/2\sec^2\theta\,d\theta}{\sqrt{9(\tan^2\theta+1)}} =$$

$$\int \frac{3/2\sec^2\theta\,d\theta}{3\sec\theta} = \frac{1}{2}\int \sec\theta\,d\theta =$$

$$\frac{1}{2}\ln|\sec\theta + \tan\theta| + c$$

$$\frac{1}{2}\ln\left|\frac{\sqrt{4x^2+9}}{3} + \frac{2x}{3}\right| + c$$

MATH 152 **QUIZ 4A** Name Solutions

Mrs. Bonny Tighe 7.2, 7.3

25 points TA_____ Wed 9/25/2019

1. Evaluate the integrals.

a) $\int \tan^4(2x)\,dx$ $\int \tan^2(2x)\tan^2(2x)\,dx$

$\int \tan^2(2x)(\sec^2 2x - 1)\,dx = \int \tan^2(2x)\sec^2(2x) - \tan^2(2x)\,dx$

4pts $\frac{1}{2}\cdot\frac{1}{3}\tan^3(2x) - \int \tan^2(2x)\,dx$

$\frac{1}{6}\tan^3(2x) - \int(\sec^2(2x)-1)\,dx$

$\frac{1}{6}\tan^3(2x) - \frac{1}{2}\tan 2x + x + C$

b) $\int \sec^4 x \tan^3 x\,dx$ $\int \sec^2 x \cdot \sec^2 x \tan^3 x\,dx$

4pts $\int(1+\tan^2 x)\tan^3 x \sec^2 x\,dx$ $du = \sec^2 x$, $u = \tan x$

$\int(\tan^3 x + \tan^5 x)\sec^2 x\,du$

$\frac{1}{4}\tan^4 x + \frac{1}{6}\tan^6 x + C$

3 pts c) $\int t\cos^2 t\,dt$

$u = t \qquad dv = \cos^2 t\,dt$

$du = dt \qquad \left(\frac{1}{2} + \frac{1}{2}\cos 2t\right)dt$

$v = \frac{1}{2}t + \frac{1}{4}\sin 2t$

$t\left(\frac{1}{2}t + \frac{1}{4}\sin 2t\right) - \int\left(\frac{1}{2}t + \frac{1}{4}\sin 2t\right)dt$

$t\left(\frac{1}{2}t + \frac{1}{4}\sin 2t\right) - \frac{1}{4}t^2 + \frac{1}{8}\cos 2t + C$

5 pts d) $\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}\,dx$

$x = 2\sin x \qquad dx = 2\cos x\,dx$

$x/2 = \sin x$

$\sqrt{3}/2 = \sin x \qquad x = \pi/3$

$1/2 = \sin x \qquad x = \pi/6$

$\int_{\pi/6}^{\pi/3} \frac{\sqrt{4-4\sin^2 x}}{4\sin^2 x}\,2\cos x\,dx = \int_{\pi/6}^{\pi/3} \frac{\sqrt{4(1-\sin^2 x)}\,2\cos x\,dx}{4\sin^2 x}$

$\int_{\pi/6}^{\pi/3} \frac{2\cos x\,2\cos x}{4\sin^2 x}\,dx = \int_{\pi/6}^{\pi/3} \frac{\cos^2 x}{\sin^2 x}\,dx$

$\int_{\pi/6}^{\pi/3} \cot^2 x\,dx = \int_{\pi/6}^{\pi/3} (\csc^2 x - 1)\,dx = -\cot x - x \Big|_{\pi/6}^{\pi/3}$

$(-\cot \pi/3 - \pi/3) - (-\cot \pi/6 - \pi/6)$

$-\sqrt{3}/3 - \pi/3 + \sqrt{3} + \pi/6$ or $\frac{2\sqrt{3}}{3} - \pi/6$

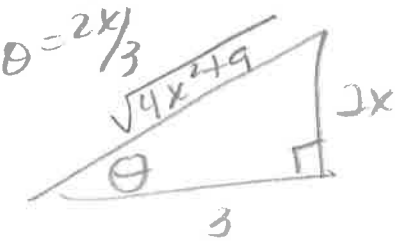
e) $\int \frac{1}{x\sqrt{4x^2+9}}dx$

5pts

$2x = 3\tan\theta \quad \tan\theta = 2x/3$

$x = 3/2 \tan\theta$

$dx = 3/2 \sec^2\theta d\theta$



$$\int \frac{3/2 \sec^2\theta d\theta}{3/2 \tan\theta \sqrt{9\tan^2\theta + 9}} = \int \frac{\sec^2\theta d\theta}{\tan\theta\sqrt{9(\tan^2\theta+1)}}$$

$$\int \frac{\sec^2\theta d\theta}{\tan\theta \, 3\sec\theta} = \frac{1}{3}\int \frac{\sec\theta}{\tan\theta}d\theta = \frac{1}{3}\int \frac{1}{\cos\theta}\cdot\frac{\cos\theta}{\sin\theta}d\theta$$

$$\frac{1}{3}\int \frac{1}{\sin\theta}d\theta = \frac{1}{3}\int \csc\theta d\theta = \frac{1}{3}\ln|\csc\theta - \cot\theta| + C$$

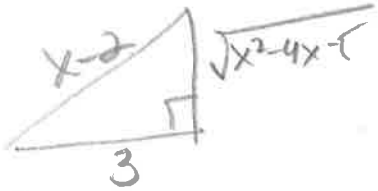
$$\frac{1}{3}\ln\left|\frac{\sqrt{4x^2+9}}{2x} - \frac{3}{2x}\right| + C$$

5pts f) $\int \frac{\sqrt{x^2-4x-5}}{x-2}dx$

$(x-2)^2 - 9$

$(x-2) = 3\sec\theta$

$dx = 3\sec\theta\tan\theta d\theta$



$$\int \frac{\sqrt{(x-2)^2-9}}{(x-2)}dx = \int \frac{\sqrt{9\sec^2\theta - 9}}{3\sec\theta} 3\sec\theta\tan\theta d\theta$$

$$\int \sqrt{9(\sec^2\theta - 1)}\tan\theta d\theta = \int 3\tan^2\theta d\theta$$

$$3\int(\sec^2\theta - 1)d\theta = 3\tan\theta - 3\theta + C$$

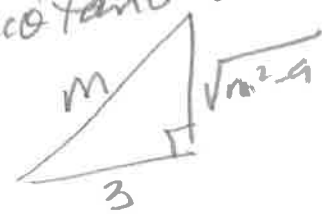
$$3\left(\frac{\sqrt{x^2-4x-5}}{3}\right) - 3\sec^{-1}\left(\frac{x-2}{3}\right) + C$$

5 pts

$(x+1)^2-9 \qquad m = x+1 \quad dm = dx$

e) $\int \frac{\sqrt{x^2+2x-8}}{x+1}\,dx \qquad \int \frac{\sqrt{m^2-9}}{m}\,dm$

$m = 3\sec\theta$
$dm = 3\sec\theta\tan\theta\,d\theta$



$\int \frac{\sqrt{9\sec^2\theta-9}}{3\sec\theta}(3\sec\theta\tan\theta\,d\theta)$

$\int \frac{\sqrt{9(\sec^2\theta-1)}\,3\cancel{\sec\theta}\tan\theta\,d\theta}{3\cancel{\sec\theta}} = \int 3\tan\theta\cdot\tan\theta\,d\theta$

$3\int \tan^2\theta\,d\theta = 3\int(\sec^2\theta-1)\,d\theta = 3\tan\theta - 3\theta + C$

$3\sqrt{\frac{m^2-9}{3}} - 3\sec^{-1}\left(\frac{m}{3}\right) + C$

$3\frac{\sqrt{x^2+2x-8}}{3} - 3\sec^{-1}\left(\frac{x+1}{3}\right) + C$

4 pts

f) $\int \tan^4(2x)\,dx = \int \tan^2(2x)\tan^2(2x)\,dx$

$\int (\sec^2(2x)-1)(\tan^2(2x)\,dx$

$\int \tan^2 2x\sec^2(2x)\,dx - \int \tan^2(2x)\,dx$

$-\int(\sec^2(2x)-1)\,dx$

$u = \tan 2x$
$du = 2\sec^2 2x\,dx$

$\frac{1}{2}\cdot\frac{1}{3}\tan^3(2x) - \frac{1}{2}\tan(2x) + x + C$

$\frac{1}{6}\tan^3(2x) - \frac{1}{2}\tan(2x) + x + C$